

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:25

Annexure A

Concept Mapping

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Concept Mapping

Question 1 (5 Marks)

Construct a multi-level concept map integrating:

Operating System Types → OS Structures → System Calls → OS Functions.

Your map must show cross-links (e.g., how OS structure influences system call implementation).

Question 2 (5 Marks)

Design a comprehensive concept map representing:

Process Management, Memory Management, File System, and I/O System.

Include interdependencies and indicate where system calls interact with each component.

Question 3 (5 Marks)

Create a comparative concept map for:

Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.

Highlight differences, similarities, advantages, and limitations using linking phrases.

Question 4 (5 Marks)

Develop a dynamic concept map showing the execution flow:

User Program → API → System Call Interface → Kernel → Hardware → Response back to User.

Include feedback loops and control flow paths.

Question 5 (5 Marks)

Construct a hierarchical + functional concept map for system calls:

Process, File, Device, and Information system calls.

Also show how they relate to OS services like scheduling, memory allocation, and device handling.

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand